

**BACHELOR OF COMPUTER SCIENCE
SCHOOL OF COMPUTER SCIENCE
BINA NUSANTARA UNIVERSITY
JAKARTA**

ASSESSMENT FORM

Course: SCIE6063001– Computational Physics

Method of Assessment: Project

Semester/Academic Year : 3/2025-2026

Name of Lecturer :

Date :

Class : All

**Topic : Voltage and Current, Resistance and Capacitance, Circuits Design & Analysis
Induction and inductance, Electromagnetic Oscillation and Alternating Current**

Group Members :	1
	2
	3
	4
	5
	6
	7
	8

Student Outcomes:

(SO 1) Mampu menganalisis masalah komputasi yang kompleks dan mengaplikasikan prinsip komputasi dan keilmuan lain yang sesuai untuk mengidentifikasi solusi.

Able to analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions

Learning Objectives:

(LObj 1.2) Mampu menerapkan prinsip komputasi dan disiplin ilmu terkait lainnya untuk mengidentifikasi solusi

Able to apply principles of computing and other relevant disciplines to identify solutions

Learning Outcomes :

(LO 1) Menjelaskan konsep dan penerapan nyata dari Listrik dan Magnet

Explain the concept and real-world applications of Electricity and Magnetism

(LO 2) Menyelesaikan permasalahan kuantitatif pada topik Listrik dan Magnet

Solve quantitative problems in topics of Electricity and Magnetism

(LO 3) Melakukan percobaan simulasi rangkaian listrik dasar menggunakan bahasa pemrograman Python

Experiment basic electrical circuits simulations using Python programming language

(LO 4) Menganalisis hasil percobaan yang direncanakan untuk menyusun laporan ilmiah

Analyze the results of the planned experiments to create scientific reports

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
2	LO1, LO2, LO3-L.Obj.1.1-SO1	Ability to solve complex problems involving of Voltage and Current, Resistance and Capacitance, Circuit, Induction, and Electromagnetic Oscillation and Alternating Current	60%	Able to design appropriate topology, assign appropriate variables and parameters, and solve the problem using computational approaches well without errors.	Able to design appropriate topology, assign appropriate variables and parameters, and solve the problem using computational approaches with minor issues	Able to design appropriate topology, but unable to assign appropriate variables and parameters, and unable to solve the problem using computational approaches with major issues	Unable to design appropriate topology, assign appropriate variables and parameters, and unable to solve the problem using computational approaches with major issues	100	60

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
3	LO4-L.Obj.1.1-SO1	Ability to solve circuit problem using Python Power Electronics to create scientific report using IMRaD structure	40%	The computation is correct without errors, and the report is completely written in IMRaD stucture	The computation can work well with some errors, but the the report is written with minor incompleteness	The computation is correct but cannot work well, and the the report is written with minor incompleteness	The computation is incorrect and cannot work well, and report is written with major incompleteness	100	40
Total Score: $\sum(\text{Score} \times \text{Weight})$									

Remarks:

ASSESSMENT METHOD

Instructions for Students:

1. Please read **PROBLEM (SECTION 2)** first before working on the other problems!
2. Submit the answer in **one .PDF document**. **DO NOT** use hand-writing in any part of your answer.
3. This case study & project are consist of several open questions, which mean that each question can be answered by using different method. You'll need to decide yourself which method is suitable for each question. You **MUST** provide the reason on choosing the method in your answer.
4. You are encouraged to provide the circuit topology, screenshot and graph from Excel or Python Power Electronics to explain your answer.
5. Your answer will be checked for similarity with other student's answer. Any similarity greater than 30% (for each answer) will result in automatic 0 (zero) points.

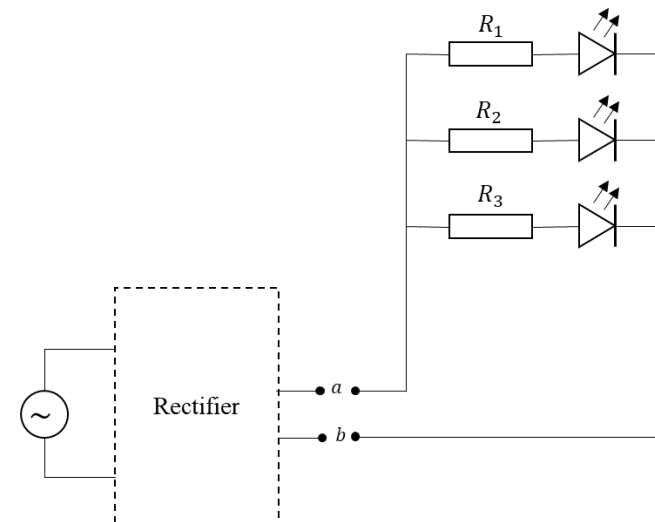
Problem (Section 1)

You are planning to conduct a small electronic project. You are planning to make a simple LED circuit (see diagram). The circuit contains three green LED lights that connected using parallel connections. From the LED specification sheet, you know that the LED will works on a minimum voltage of 2 V . Also, you gain information that the LED will break if the current that flow through it exceed 20 mA . For that, you need to use some resistor to limit the current that flow through the LED.

To power the circuit, you are planning to use a micro hydro generator that you already have. However, you realize that you need to convert the current from AC to DC so you can light up the LED. But you don't have the proper converter at the moment. So, you decide to make your own rectifier, *a basic RLC rectifier*, from only the components you have (see table) at hand. Assume that your component supply is large enough, so you can use any number of each component.

Assume that your LED does not have any internal resistance. The LED circuit will be connected to the rectifier at point *a* and *b* (see diagram). The generator has an output of 5 V with frequency of 20 Hz . **So, what is your solution for the basic RLC rectifier?** Assume that each component on the table (including the EMF) has internal resistance of $0.1\ \Omega$.

No	Component	Value
1	Resistor	$3\ \Omega$
2		$24\ \Omega$
3		$36\ \Omega$
4		$100\ \Omega$
5		$130\ \Omega$
6		$220\ \Omega$
7		$510\ \Omega$
8	Capacitor	$1.0\ \mu\text{F}$
9		$3.3\ \mu\text{F}$
10		$22\ \mu\text{F}$
11	Inductor	$2\ \mu\text{H}$
12		$5.1\ \mu\text{H}$
13		2 mH
14		400 mH



Problem (Section 2)

Prepare all your answers for Problem (Section 1) in the form of a scientific report in **maximum of 2 sheets of paper (1-sided)**. Your lecturer will evaluate the clarity, completeness, and scientific quality of your report. The report should include the following sections using IMRaD structure :

- a. Title and short Abstract
- b. Introduction (background and objectives)
- c. Methods (explanation of models, numerical approaches, and code used)
- d. Results and Discussion (analysis of the obtained results, including graphs, tables, and error analysis if necessary)
- e. References (if any)

Note for Lecturers:

1. The lecturers are advised to assess student's understanding towards the topics included in the assignment: Voltage and Current, Resistance and Capacitance, Circuits Design & Analysis, Induction and inductance, Electromagnetic Oscillation and Alternating Current.
2. The students will submit their answer in .PDF format through BINUS Maya.
3. The deadline of this comprehensive assignment is at the end the semester.